RnD Report
On

FORECASTING PM2.5 CONCENTRATIONS USING FACEBOOK PROPHET AND LSTM

by

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DECLARATION

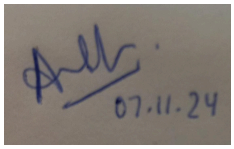
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ABSTRACT

The forecasting of PM2.5 concentration is of utmost importance for environmental policymakers and human health, since long inhalation of high concentrations of the particulate matter can lead to serious respiratory and cardiovascular diseases. There are many available forecasting technologies which include statistical and machine learning techniques, each having their advantages and challenges. The objective of this project is to review existing work in this field and then employ Facebook Prophet and LSTM models to forecast PM2.5 concentrations and analyze the results. First, a thorough conceptual review and model comparison were carried out for ARIMA, Facebook Prophet, LSTM, and GRU models to understand their working mechanisms and choose the best models for the scope of this study. Next, a literature review of the already conducted studies showed how the models were constructed and the kind of results that were obtained, thus informing this study's methodology. While utilizing Central Pollution Control Board (CPCB) daily data from October 2021 to October 2024, the models were built by training them on the first two years and the PM2.5 concentration levels were forecasted for the third year. The evaluation was conducted using RMSE and R-squared measures. The project's findings contribute to more informed model selection and improved air quality forecasting practices.

Keywords: PM2.5, Machine learning, Facebook Prophet, LSTM

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Chapter 1

Introduction

Air pollution has emerged as one of the major environmental challenges across the globe. It causes severe health problems and is responsible for many premature deaths. Research in India shows that almost 40% of the population will see a decline in life expectancy due to bad air quality (Lee & Greenstone, 2021). Fighting air pollution calls for a dire need to develop an efficient forecasting model that can predict the concentration of pollutants like PM2.5 to reduce the health impacts related to it.

Air quality forecasting is a very complex task, and requires an understanding of the interaction between different types of pollutants and meteorological factors that influence them. The traditional methods of forecasting like statistical models, ARIMA have been broadly used as they are very simple and rely on historical data. However, its inability to capture nonlinear relationships and dependency on stationary data limit its effectiveness in complex environmental settings. The need for advanced predictive models that can surpass these limitations and provide more accurate forecasts is of utmost importance.

The recent development in machine learning (ML) has created new opportunities for air quality forecasting. Techniques in ML are noted to find intricate patterns and relationships on huge datasets. This puts them at an advantage over greater reliability predictions. Out of these, two promising approaches for time-series forecasting are LSTM networks and Facebook Prophet. LSTM networks are a specific class of recurrent neural networks. These networks model sequential data, which makes them suitable for time series analysis, for example, predicting air quality levels based on historical data. LSTMs can learn long-term dependencies in the data so that they can capture trends and seasonality that traditional models miss.

On the other hand, Facebook Prophet is user-friendly, particularly built for forecasting time series data, accounting for missing values and outliers. It automatically fits seasonality and holiday effects. Prophet, being flexible and user-friendly, forms a rich tool in the toolbox for environmental scientists and policymakers for effectively forecasting air quality.

The objective of this project is to research the potential for forecasting concentrations of PM2.5 by using data received from the Central Pollution Control Board (CPCB), India through LSTM and Facebook Prophet. We train these models on data for the first two years: October 2021 to October 2023. They predict the PM2.5 concentrations for the subsequent year, October 2023 to October 2024.

This research also focuses on data preprocessing and feature engineering apart from model implementation. An in-depth understanding of what factors affect pollutant levels is necessary for accurate air quality forecasting, as this would include meteorological factors like temperature and humidity. Incorporating these variables into the models should help improve their performance as predictors and build a much stronger framework for forecasting.

Through using machine learning techniques, the research will add value to the body of studies about air quality forecasting. The demonstration that it is possible to predict PM2.5 concentration with LSTM and Facebook Prophet should thereby establish a foundation for further future progress in the development of an air quality management strategy. Better decision-making processes may be obtained with the successful application of these models, and policymakers and environmental agencies will be able to develop targeted interventions to protect public health and improve air quality. We aim through this research to make a meaningful contribution to the ongoing efforts to combat air pollution and its associated risks, towards fostering a healthier and more sustainable future for all

Chapter 2

Conceptual Analysis and Model Comparison

In predictive modeling, it is crucial to understand how each model works, thus guiding the decision about which model to apply. Since machine learning models could often produce results very close to the true ones, treating them as "black boxes" with little knowledge of their functioning limits their applicability and interpretation, especially when used for sensitive applications like forecasting air quality. Each model will have different assumptions, strengths, and limitations that can affect its performance based on the characteristics of the data, the forecasting horizon, and the underlying patterns in the data. Thus, it is necessary to conduct a conceptual analysis to identify the appropriate model for this study.

This section elaborates upon the theoretical working of four dominant models that can be used in time series forecasting: ARIMA, Facebook Prophet, LSTM, and GRU. Also, a comparative analysis has been done to justify why LSTM and Facebook Prophet have been chosen for this work.

2.1 ARIMA Model (AutoRegressive Integrated Moving Average)

The ARIMA model is a widely used statistical method for time series forecasting, particularly effective for short-term predictions of data that exhibit linear trends and stationary behavior. ARIMA incorporates three main components:

- **Autoregressive (AR) Component**: Uses past values of the time series to predict future values, assuming that previous observations are linearly correlated with future values.
- **Integrated (I) Component**: Differencing is applied to make the data stationary, which removes trends or seasonality. This is a necessary step for ARIMA as it assumes constant mean and variance, although differencing can sometimes lead to data loss.
- **Moving Average (MA) Component**: Models past errors in the time series, which helps smooth out fluctuations and correct short-term variations in the data.

The ARIMA model is expressed mathematically as:

$$Y_t = (\beta_1 + \beta_2) + (\Phi_1 Y_{t-1} + \cdots + \Phi_p Y_{t-p}) + (\omega_1 \varepsilon_{t-1} + \cdots + \omega_q \varepsilon_{t-q} + \varepsilon_t)$$

Fig 2.1: Mathematical expression of ARIMA model

where Yt represents the predicted value, and the bracketed terms represent intercepts, lagged values, and lagged errors in order, with w and phi representing coefficients.

While ARIMA is effective for short-term linear patterns and stationary data, it has significant limitations. It assumes a linear relationship and requires preprocessing to achieve stationarity, which can cause loss of valuable information. Additionally, it does not account for external variables or nonlinear interactions, making it less suitable for complex data such as air quality influenced by external factors like weather.

2.2 Facebook Prophet

Facebook Prophet is an open-source forecasting model designed to handle time series data with strong seasonality, trends, and holiday effects. Prophet decomposes time series data into three components:

- **Trend Component**: Captures the general direction of the data over time
- **Seasonality Component**: Models recurring patterns (e.g., weekly, monthly, or yearly cycles), which is particularly useful for pollution data that fluctuates seasonally
- **Holiday Component**: Allows the inclusion of specific events, such as holidays, where spikes in data occur, such as higher pollution levels during festivals

The model formula is:

$$Y(t)=g(t)+s(t)+h(t)+e$$

where Y(t) is the forecasted value, g(t) represents the trend, s(t) models seasonality, h(t) represents holiday effects, and e accounts for noise.

Facebook Prophet is advantageous for data with distinct seasonal and event-driven patterns, and it is resilient to missing data and outliers. However, it has limited support for including additional external factors (e.g., temperature, humidity) and may struggle with highly volatile data, relying mainly on historical patterns rather than complex interactions.

2.3 LSTM (Long Short-Term Memory)

LSTM networks, a type of recurrent neural network (RNN), are built to capture long-term dependencies in sequential data. LSTM models incorporate three types of gates to manage the flow of information over time:

- **Forget Gate**: Controls what information from the past is discarded, ensuring only relevant information is retained.
- **Input Gate**: Decides what new information to add to the memory cell, helping update predictions based on both recent and long-term trends.
- **Output Gate**: Determines what part of the memory is output at each time step, affecting the forecasted value.

The cell state in LSTM serves as a memory unit that is updated through these gates, allowing the model to learn complex and nonlinear dependencies over time.

LSTM is highly suitable for time series with nonlinear patterns and long-term dependencies, such as air quality data influenced by various factors over time. However, it is computationally intensive, requiring significant resources and data for training. Additionally, LSTM's internal mechanisms are complex and less interpretable.

2.4 GRU (Gated Recurrent Unit)

GRU is a simplified version of LSTM, designed to retain many of LSTM's benefits while reducing complexity. GRU operates with two gates:

- **Reset Gate**: Determines how much of the previous hidden state to forget, balancing past and new information.
- **Update Gate**: Controls how much new information to add to the current hidden state, enabling a balance between retaining historical data and incorporating new input.

GRU's simpler architecture allows it to achieve comparable performance to LSTM but with reduced computational costs and faster training times. It is beneficial for applications requiring efficient processing, as it balances performance and training speed. However, it may lack the depth of LSTM in capturing highly complex and intricate patterns in data. Its simpler design also provides limited flexibility, making it less suitable for datasets with irregular patterns.

LSTM was chosen for its ability to capture complex, nonlinear relationships in time series data and manage long-term dependencies. This capability is crucial for air quality forecasting, where pollutant levels are influenced by various dynamic factors over extended periods.

Facebook Prophet was selected for its robustness in capturing seasonality and holiday effects, both of which are prevalent in air quality data, especially in regions with distinct pollution spikes around specific events or times of year.

Chapter 3

Literature Review of Existing Methods to Forecast PM2.5 Values

3.1 Air Quality Index Forecasting Using Parallel Dense Neural Network and LSTM Cell

This study used a hybrid approach with LSTM and DNN to enhance the prediction of air quality with the inclusion of meteorological factors separately. This was a PM2.5 forecasting model, based on data from Beijing and all associated meteorological factors such as temperature, wind speed, and pressure. Meteorological data, separate from the PM2.5 sequences, allowed DNN to process external information independently without carryover memory effects, as might occur with LSTM. The hybrid model was fit over 40,000 iterations with 60-day batches of Huber loss and Adam optimizer. Results showed that, although LSTM alone performed well for short-term predictions, the hybrid LSTM-DNN model also outperformed it in extended forecasting periods because of the added impact of meteorological factors.

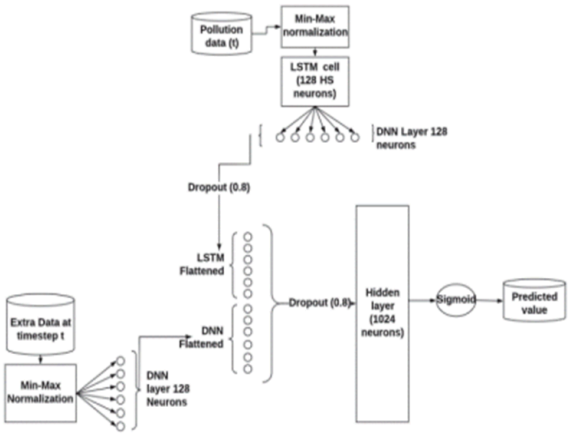


Fig 3.1 Model Architecture

Sr. No	Hybrid Model	Simple LSTM
1	36.426	61.795
2	68.593	75.087
3	26.436	34.060

Table II. MAE values of test 2 for 16 days forecast

Sr. No	Hybrid Model	Simple LSTM
1	32.783	58.634
2	81.193	90.360
3	28.535	70.098

Table III. MAE values of test 2 for 30 days forecast

Fig 3.2 Model Results

3.2 Applying Machine Learning Techniques in Air Quality Prediction

This study evaluated the performance of ARIMA, Facebook Prophet, and LSTM for the prediction of PM2.5 concentrations across seven Russian cities using data from the World Air Quality Index platform. The dataset included pollutant concentrations (PM2.5, PM10, ozone) and meteorological data from 2014 to 2021. Prophet was tuned

to capture the daily, weekly, and yearly trends that would easily capture the seasonal variation in air quality. In the end, LSTM configured with 100 units and ReLU activation outperformed both Prophet and ARIMA by performing better on MSE and RMSE. The complexity of modeling long nonlinear patterns explained why LSTM excelled in capturing pollutant variability compared with ARIMA or Prophet, which were more suitable for short- and medium-term patterns rather than long-term trends.

Models	RMSE Errors	MSE Errors
ARIMA	12.459111	155.229452
Prophet	12.247773	150.007950
LSTM	7.865484	61.865833

Fig 3.3 Paper 2 Results

3.3 Air Quality Forecasting Using LSTM RNN and Wireless Sensor Networks

The model is a combination of LSTM with data from WSN to carry out real-time air-quality forecasting across Bengaluru, India. It received pollutant data for PM2.5, PM10, SO_2, NO_2, O_3, CO, and NH_3 at minutely intervals and added data at hourly intervals from the government through a Data API. LSTM has been used to capture time-dependent patterns in pollutant concentration. Nevertheless, the error spread across the cities with RMSE being higher in the case of Bengaluru compared to that in Amravati. Interestingly, it did not include meteorological variables like wind speed and humidity which may have caused an error in the prediction levels of places with varied patterns. The authors proposed that the inclusion of these variables in the model would increase the accuracy of the prediction.

These studies reflect the superiority of both approaches to predicting air quality values - each fits different necessities and dimensions of complexity presented in the data. LSTM does better in catching strong nonlinear interplays and elaborate dependencies by time, significantly on hybrid combinations, separately encompassing external aspects. Whereas, in particular data featuring marked seasonality, the Facebook Prophet turns out quite useful due to the successful decomposing between trends and features of seasonality. These results justify LSTM's use for processing complex dependencies in PM2.5 forecasts, though Prophet's feature of seasonal modeling does provide an alternative efficient method to extract periodic patterns.

Chapter 4

Models and Results

4.1 Data Preprocessing

The daily air quality data of the Patparganj station in Delhi, collected by the Central Pollution Control Board (CPCB), was used for this study. It encompasses a period from October 9, 2021, to October 9, 2024, with 1,097 rows. This dataset comprises daily readings of PM2.5 concentration along with temperature and relative humidity values.

Two solutions were considered to deal with missing values: dropping the rows containing the missing values, and imputation, in which missing values are filled up by replacement of the mean value from the dataset. With a mean of only 7 missing rows in the whole set, it did not greatly affect dropping them compared with mean imputation since introducing artificial smoothing would influence data. The resulting dataset, without the 7 missing rows, was used for all further analysis and modeling.

Seasonal application of the Z-score and Interquartile Range (IQR) methods was used to capture seasonal outliers in the data, where one spiked uniquely to distort the model results. This method was more outlier-sensitive. However, both methods showed an insignificant number of outliers in the seasonal partitions, and therefore, we kept the dataset as it was with dropped rows without making further adjustments for the remainder of the analysis.

```
0 outliers removed in season monsoon using Z-Score method.
3 outliers removed in season winter using Z-Score method.
0 outliers removed in season summer using Z-Score method.
Final Z-score data saved as 'final_zscore.csv'.
0 outliers removed in season monsoon using IQR method.
0 outliers removed in season winter using IQR method.
0 outliers removed in season summer using IQR method.
Final IQR data saved as 'final_iqr.csv'.
```

Fig 4.1. Outlier detection results

4.2 Facebook Prophet: Implementation and Results

We used Facebook Prophet in this study as one of the models because PM2.5 concentrations are prone to seasonality and occasional event-driven spikes (like Diwali). Three configurations of the dataset were used for training Prophets to assess the extent to which additional meteorological variables improve the accuracy of the forecast. It was first trained using only the PM2.5 concentration and then further trained by the combination of PM2.5 concentration and temperature and finally using the combination of PM2.5, temperature, and relative humidity. To tune hyperparameters, for seasonality prior scale and changepoint prior scale, a grid search was employed for enhanced performance in each model.

Hyperparameter Tuning

The seasonality prior scale controls how flexible the seasonal component of the model is. A higher seasonality prior scale allows Prophet to closely fit seasonal fluctuations and capture

very distinct seasonal patterns. A lower value constrains the model to a smoother, averaged seasonal trend, reducing sensitivity to smaller fluctuations.

The changepoint prior scale lets the model adapt much more freely to sudden changes in trend, which can be useful for data including many sudden changes. A small value smooths the trend curve and makes it less sensitive to sharp fluctuations. This would be useful if the underlying trend is expected to be relatively stable.

Model Configurations and Testing

The data has been divided into training data and test data: October 9, 2021 to October 9, 2023 data have been used for model training, and the same model has been evaluated for performance on the test data of October 9, 2023 to October 9, 2024.

In addition to the overall estimate, seasonal analyses were performed to see if the model resolves the seasonal variability in air pollution. The partitions of seasons were defined as follows:

Winter: October to February

Summer: March to June

Monsoon: July to September

For all seasons, different models were trained for every configuration to test the seasonal dynamics' effect on model accuracy.

The trend, weekly, yearly, and extra regressors graphs illustrate different components of the model's predictions. The trend graph shows the overall pattern of the data over time, capturing long-term growth or decline, while the weekly graph displays periodic fluctuations occurring weekly, indicating any repeating weekly cycles. The yearly graph reveals seasonal effects that repeat annually, highlighting trends like yearly weather patterns or holidays. Extra regressors (if used) represent the influence of additional external factors, like temperature and relative humidity, on the predictions.

4.2.1 Overall Results:

4.2.1.1 PM2.5 only

Changepoint prior scale = 0.001

Seasonality prior scale = 0.0001

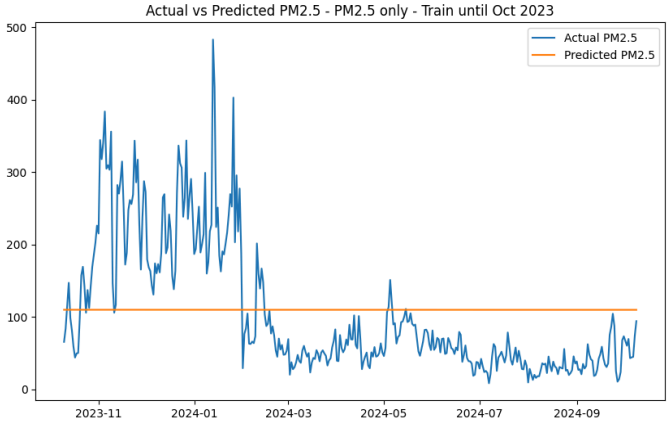


Fig 4.2. Overall: Actual vs predicted PM2.5, PM2.5 only

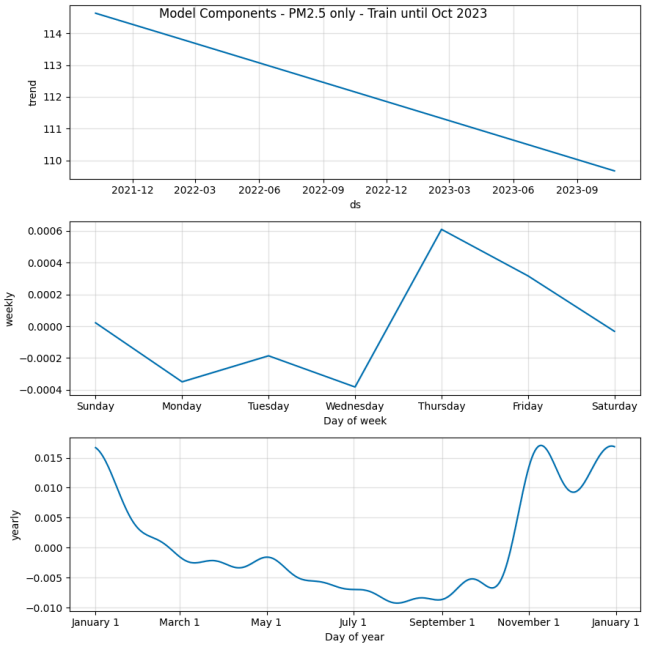


Fig 4.3. Overall: Components, PM2.5 only

4.2.1.2 PM2.5 + Temperature

Changepoint prior scale = 0.001
Seasonality prior scale = 0.0001

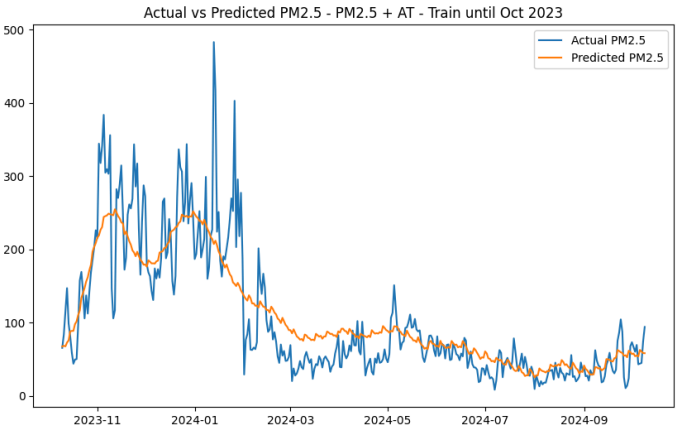


Fig 4.4 Overall: Actual vs predicted PM2.5, PM2.5 + Temperature

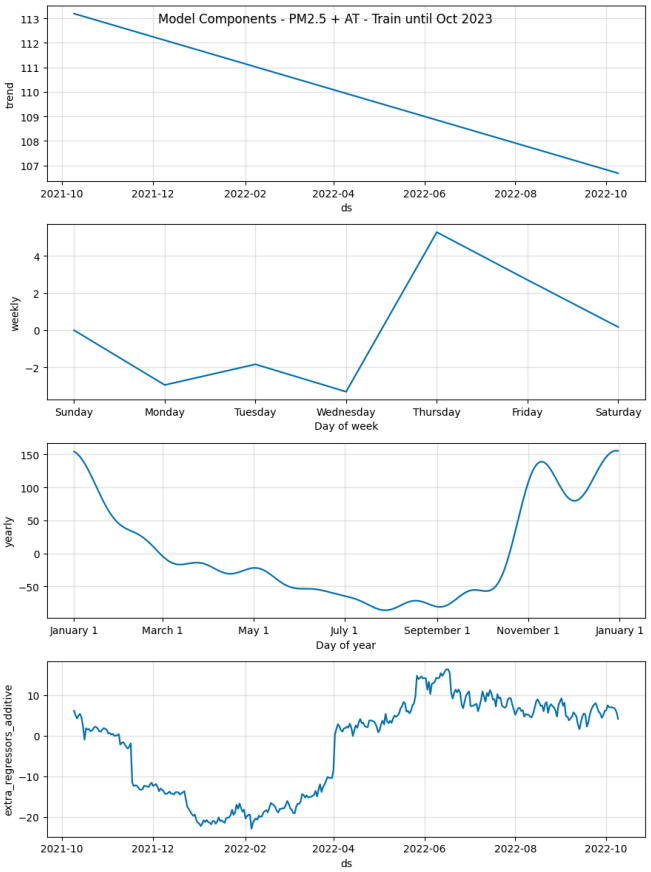


Fig 4.5 Overall: Components, PM2.5 + Temp

4.2.1.3 PM2.5 + Temperature + Relative Humidity

Changepoint prior scale = 0.001
Seasonality prior scale = 0.0001

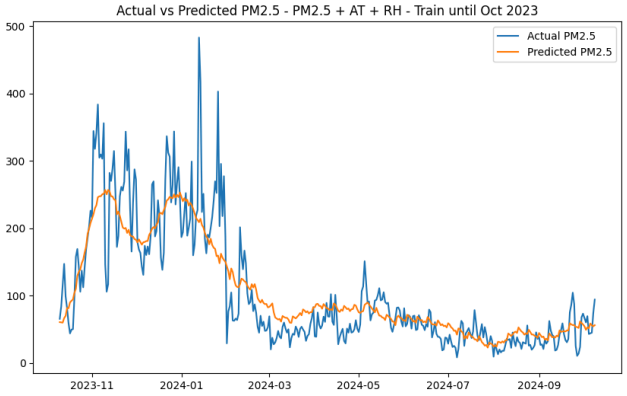


Fig 4.6 Overall: Actual vs predicted PM2.5, PM2.5 + Temperature + Relative Humidity

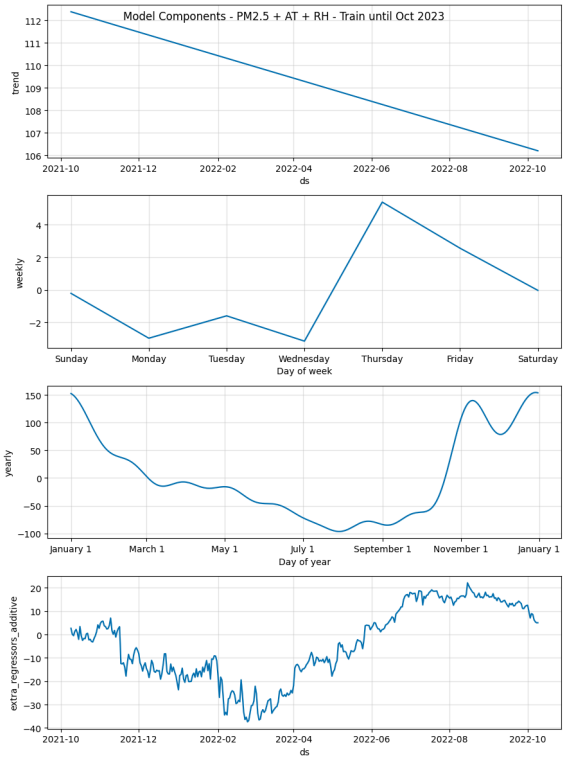


Fig 4.7 Overall: Components, PM2.5 + Temperature + Relative Humidity

	RMSE	R^2
PM2.5 only	91.76	-0.00199
PM2.5 + Temperature	46.88	0.7384
PM2.5 + Temperature + Relative Humidity	45.19	0.7569

Table 4.1 Facebook Prophet: Overall Results

We see that the model does not perform well when the meteorological factors are not considered, which could be because complex relationships have not been taken into account. However the performance consistently improves to a good R^2 score of 0.75 on adding the factors. This testifies to the importance of the factors in forecasting PM2.5 values.

4.2.2 Seasonal Results:

4.2.2.1 Winter

PM2.5 only

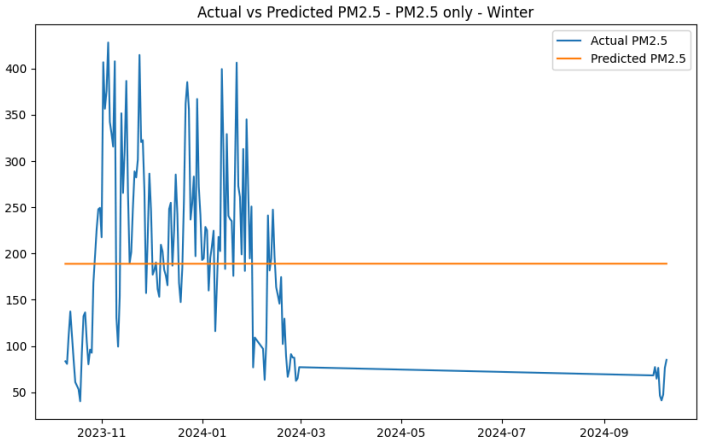


Fig 4.8 Winter: Actual vs Predicted PM2.5, PM2.5 only

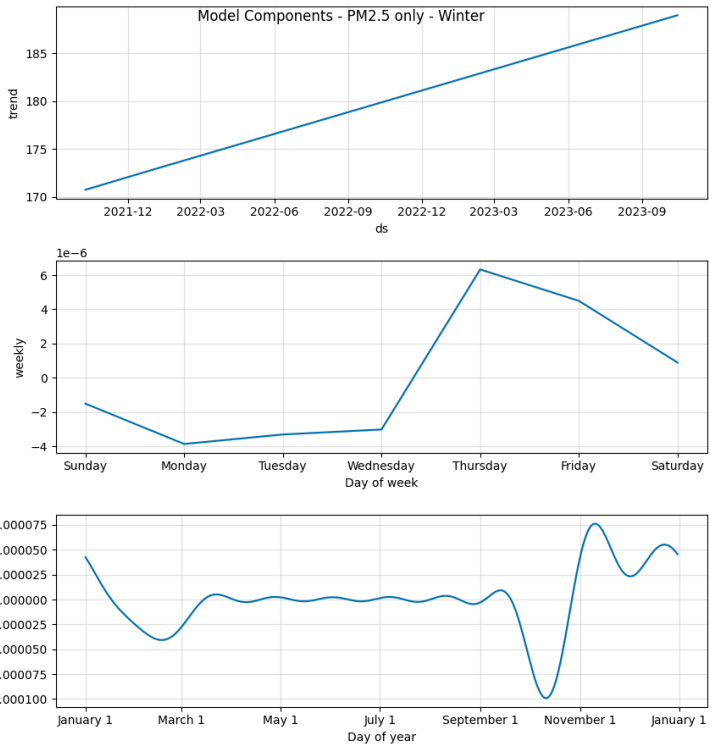


Fig 4.9 Winter: Components, PM2.5 only

PM2.5 + Temperature

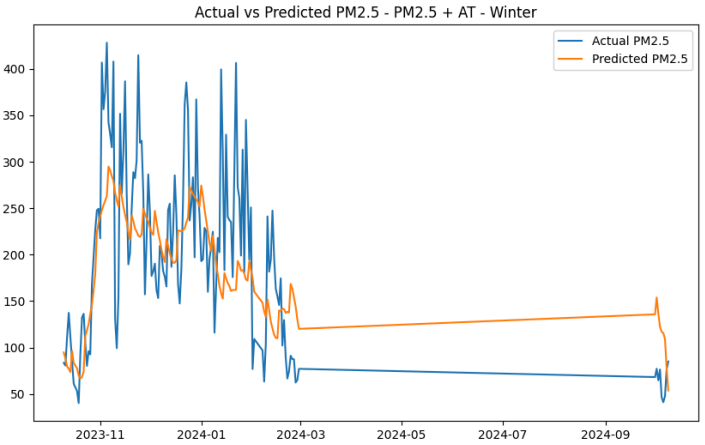


Fig 4.10 Winter: Actual vs Predicted PM2.5, PM2.5 + Temperature

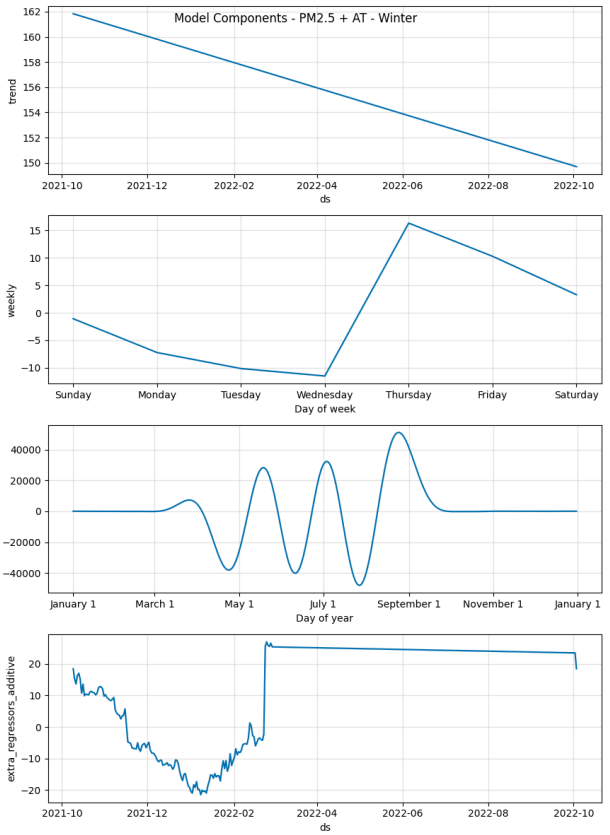


Fig 4.11 Winter: Components, PM2.5 + Temp

PM2.5 + Temperature + Relative Humidity

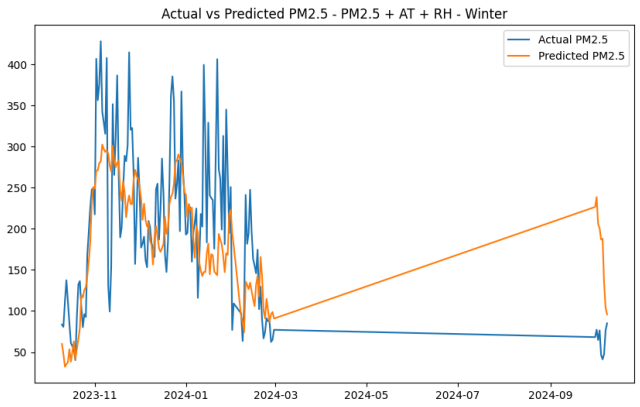


Fig 4.12 Winter: Actual vs Predicted PM2.5, PM2.5 + Temperature + RH

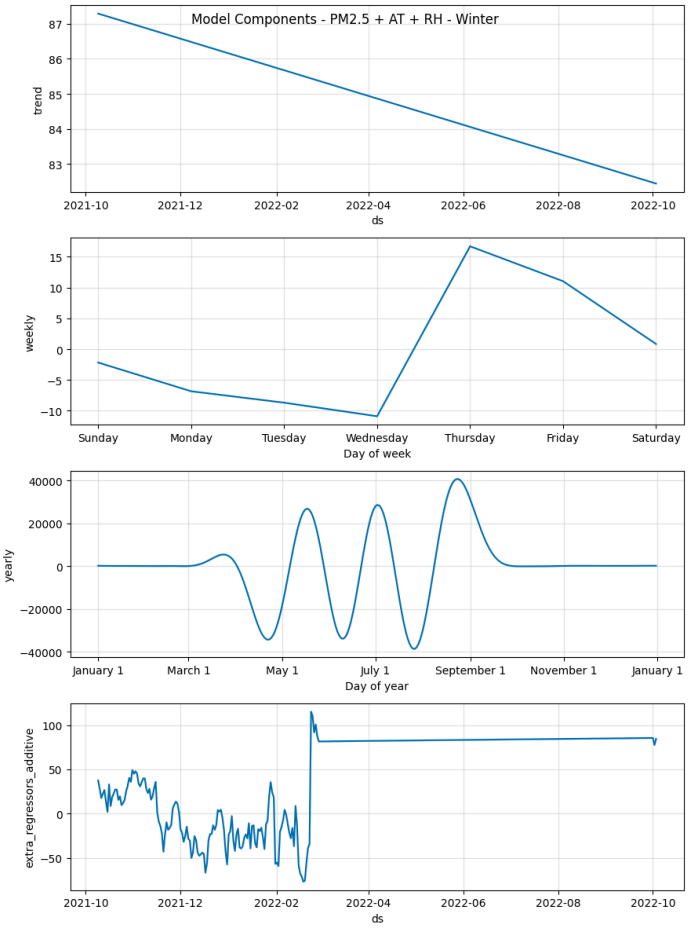


Fig 4.13 Winter: Components, PM2.5 + Temp + RH

3.2.2.2 Monsoon

PM2.5 only

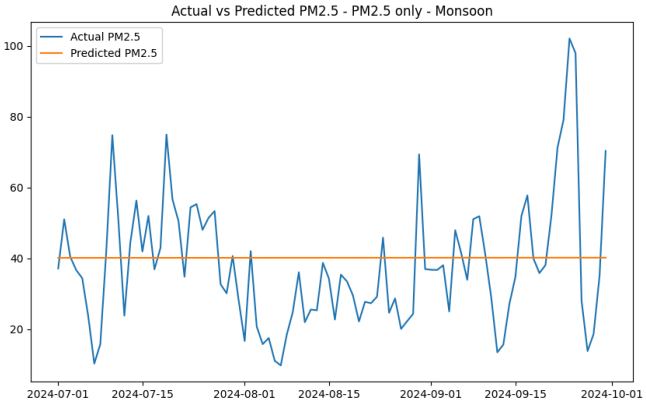


Fig 4.14 Monsoon: Actual vs predicted PM2.5, PM2.5 only

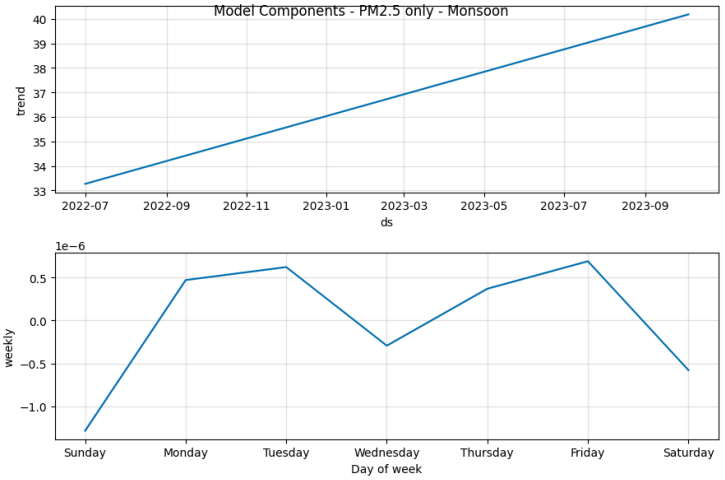


Fig 4.15 Monsoon: Components, PM2.5 only

PM2.5 + Temperature

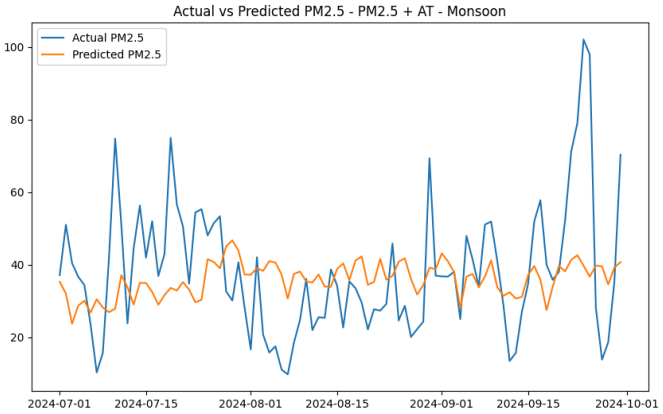


Fig 4.16 Monsoon: Actual vs predicted PM2.5, PM2.5 + Temperature

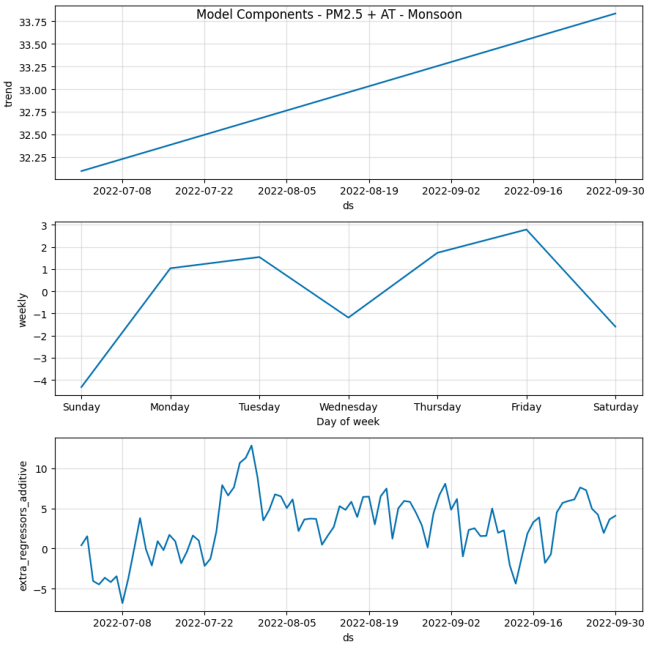


Fig 4.17 Monsoon: Components, PM2.5 + Temp

PM2.5 + Temperature + Relative Humidity

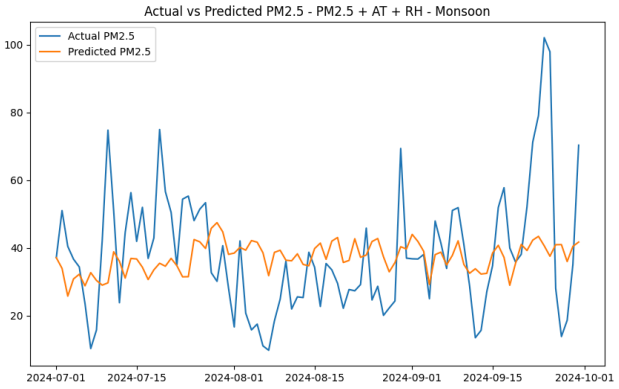


Fig 4.18 Monsoon: Actual vs predicted PM2.5, PM2.5 + Temp + RH

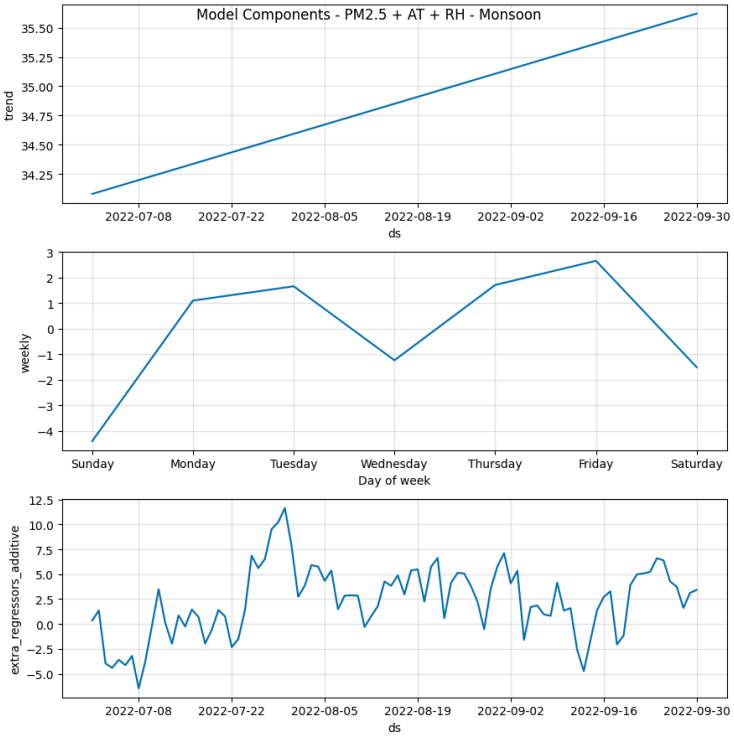


Fig 4.19 Monsoon: Components, PM2.5 + Temp + RH

3.2.2.3 Summer

PM2.5 only

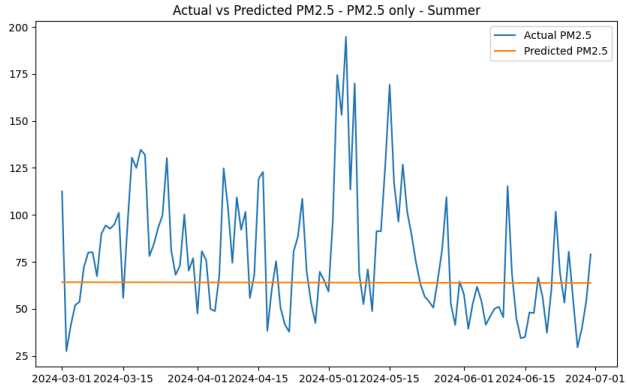


Fig 4.20 Summer: Actual vs predicted PM2.5, PM2.5 only

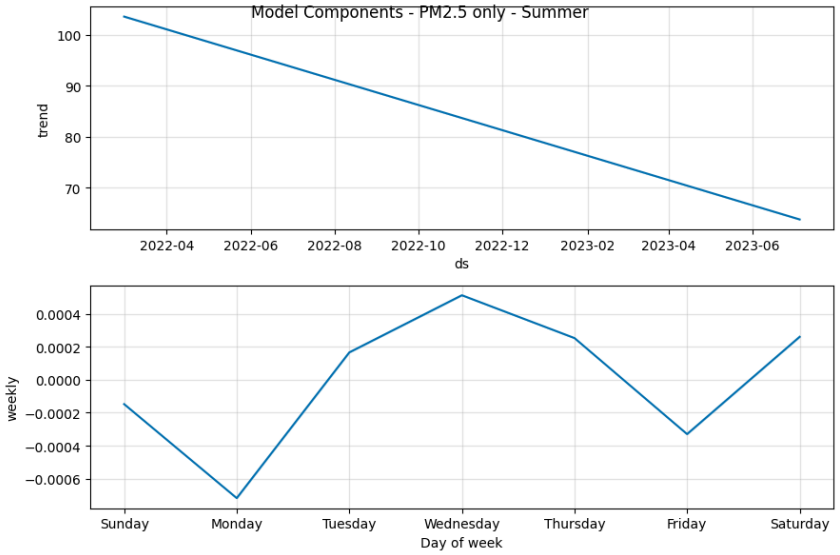


Fig 4.21 Summer: Components, PM2.5 only

PM2.5 + Temperature

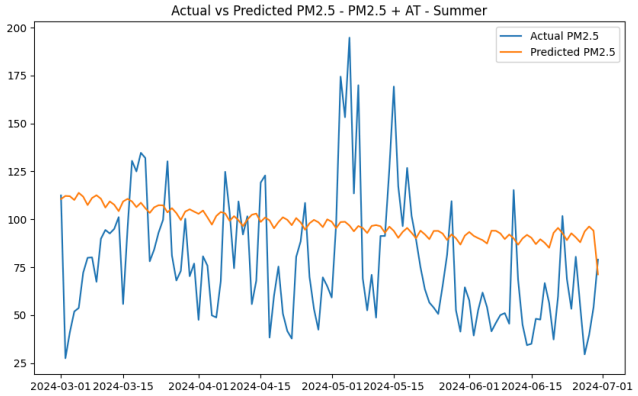


Fig 4.22 Summer: Actual vs predicted PM2.5, PM2.5 + Temp

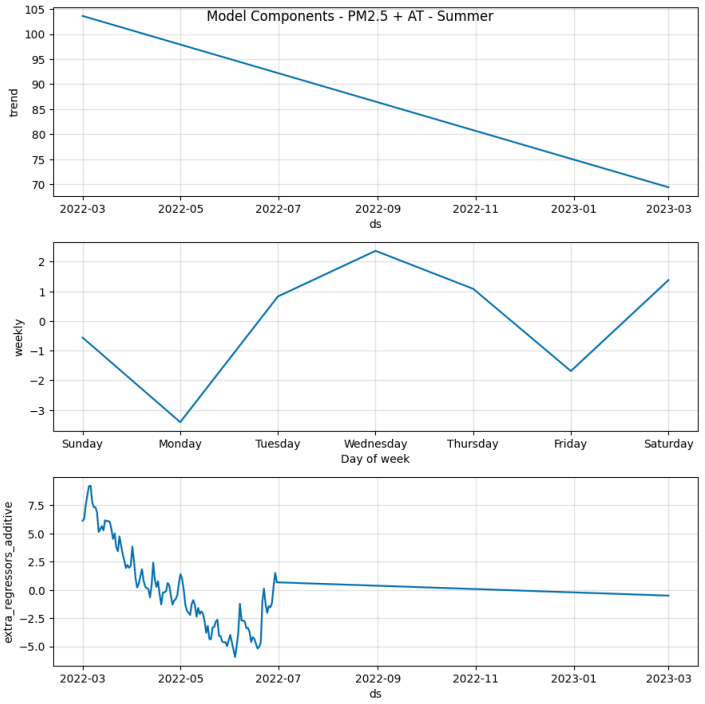


Fig 4.23 Summer: Components, PM2.5 + Temp

PM2.5 + Temperature + Relative Humidity

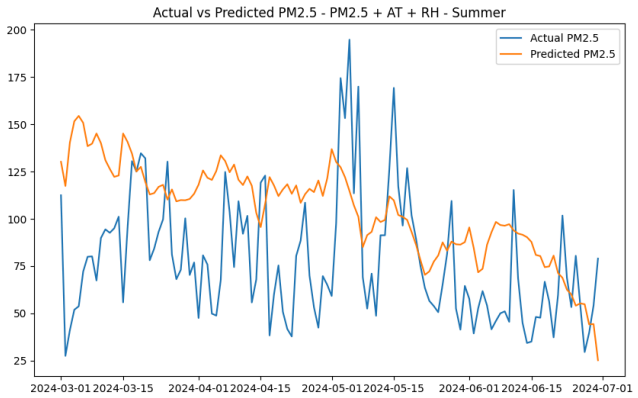


Fig 4.24 Summer: Actual vs Predicted PM2.5, PM2.5 + Temp + RH

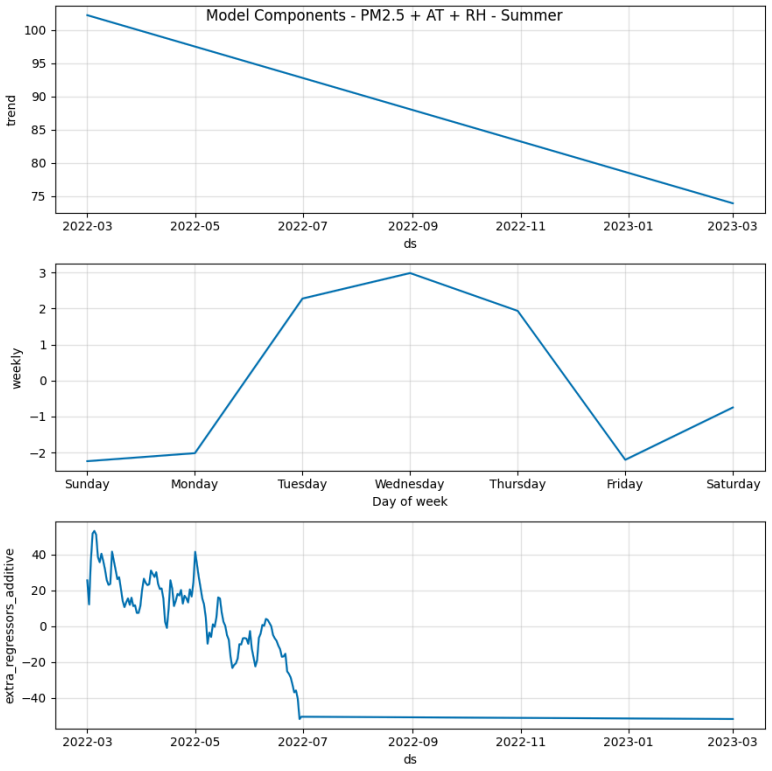


Fig 4.25 Summer: Components, PM2.5 + Temp

	RMSE	R^2	Changepoint prior scale	Seasonality prior scale
Winter				
PM2.5 only	98.15	-0.013	0.007	0.00001
PM2.5 + Temperature	78.62	0.398	0.001	50
PM2.5 + Temperature + Relative Humidity	78.3	0.354	0.001	50
Monsoon				
PM2.5 only	18.12	-0.014	0.008	0.00001
PM2.5 + Temperature	18.29	-0.032	0.02	0.2
PM2.5 + Temperature + Relative Humidity	18.14	-0.015	0.0001	50
Summer				
PM2.5 only	35.69	-0.18	0.002	0.0002
PM2.5 + Temperature	37.99	-0.337	0.0001	50
PM2.5 + Temperature + Relative Humidity	44.34	-0.821	0.0001	50

Table 4.2 Facebook Prophet: Season-wise Results and Hyperparameter Values

While the addition of meteorological factors in the model improves its performance for winter data, no clear trend can be observed for monsoon and summer data, with the model performing very poorly here. Possible reasons for this result are discussed in Chapter 5.

3.2 LSTM: Implementation and Results

Utilizing a Long Short-Term Memory (LSTM) model can capture the inherent time dependencies and complex, nonlinear relationships in the air quality data to forecast PM2.5.

Base Architecture and Hyperparameters

Here, a two-layer LSTM network architecture with dropout layers has been used to help prevent overfitting, and the final prediction from the dense output layer has been utilized.

- The first layer of this model is an LSTM layer with 150 units and ReLU activation.
- Dropout with probability 0.2. This will prevent overfitting by randomly setting the output of 20 percent of the units to zero during each training iteration.
- The second LSTM layer has 150 units, applying the ReLU activation function for sequence information coming from the preceding layer.
- Another dropout layer of 0.2 is used.
- Output Layer is a dense layer consisting of one neuron that merely gives out the final forecasted value of PM2.5.

The model was compiled with the Adam optimizer using a learning rate of 0.01, and loss was set to MSE.

The configuration used during the training of this model:
Epochs: 150
Batch Size: 32
A grid search fine-tuned hyperparameters to maximize performance over the best number of LSTM units, dropout rate, learning rate, and batch size.

Results and Analysis

RMSE: 51.389
R^2: 0.68584
The model performs fairly well giving an R^2 value of close to 0.7.

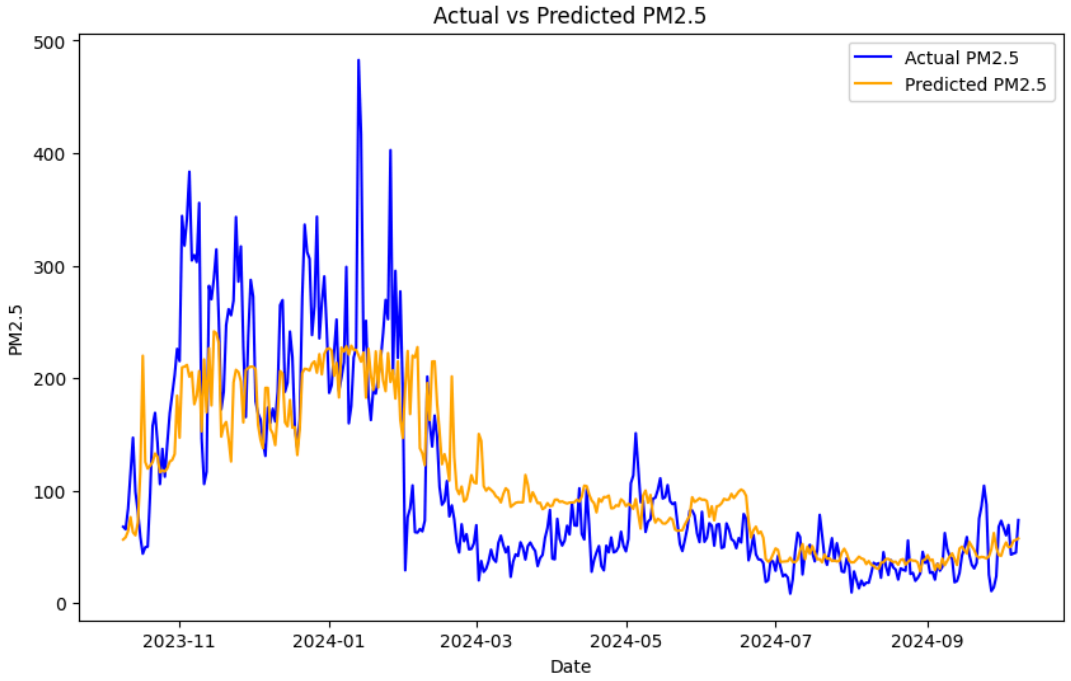


Fig 4.26 LSTM Results

Chapter 5

Discussion, Conclusion and Next Steps

In this work, we implement and compare the performance of LSTM and Facebook Prophet models in the PM2.5 concentration prediction task. The R^2 value of the LSTM model is approximately 0.7, meaning that the complex temporal dependencies and nonlinear patterns were captured from the air quality data. Facebook Prophet also performed quite well in the overall context, as indicated by the R^2 score of around 0.75, which reveals that it captures general trends and seasonality in the data set. However, the seasonal forecasts of Prophet did not perform well and missed the seasonal changes.

The training and test data are not uniform in terms of seasons, but more uniform overall. The training data was based on PM2.5 values for the first two years, whereas the third year had a different seasonal pattern, as indicated by seasonal plots with high variations between years.

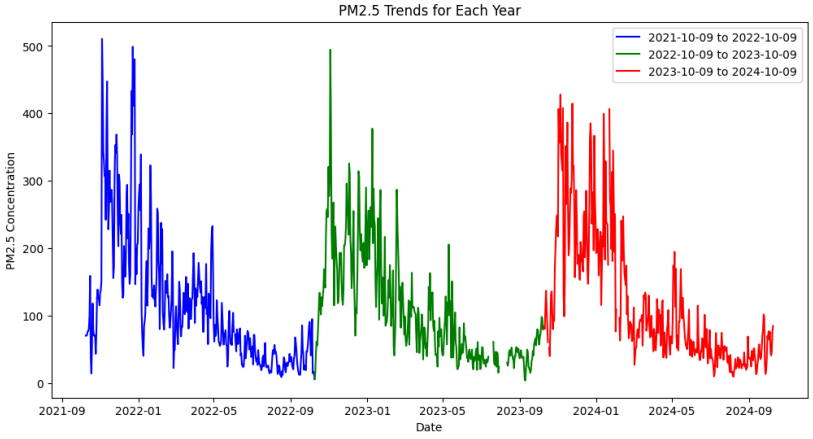


Fig 5.1 Data distribution over 3 years

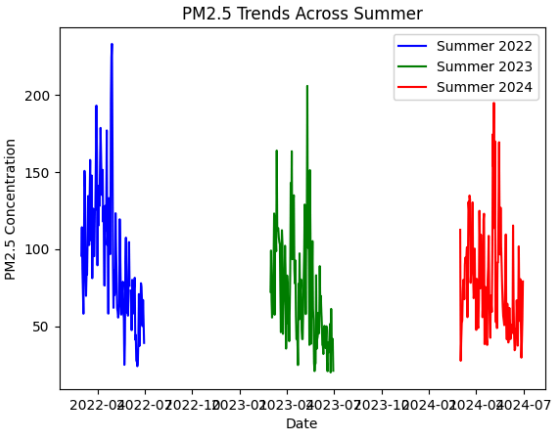


Fig 5.2 Summer data distribution

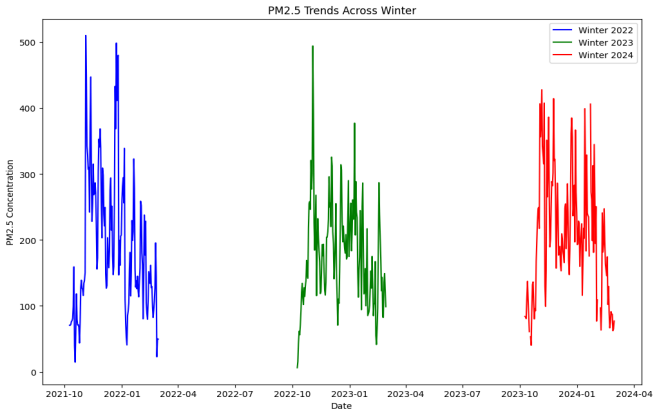


Fig 5.3 Winter data distribution

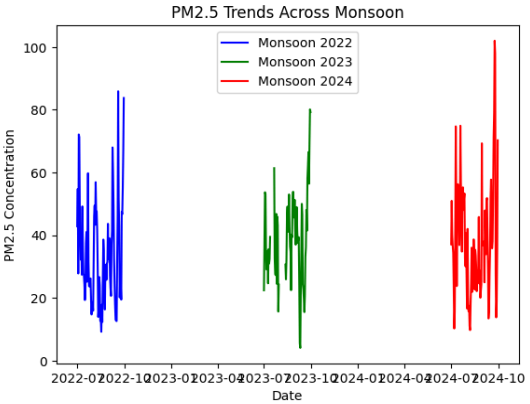


Fig 5.4 Monsoon data distribution

We tested the models again using the data from DTU in Delhi to gauge whether the limitations of good results lie with the data rather than elsewhere, but the results were similar, suggesting the necessity of further work on model architecture to improve seasonal results.

Next Steps

Include Other Meteorological Variables: Other variables like humidity, wind speed, and temperature could be included in the Prophet model as external regressors to enable the model to better account for environmental impacts with seasonal characteristics.

Enhance the quality of data: Leveraging data with fewer inconsistencies, such as from well-calibrated sensors or standardized sources, can reduce noise and enable Prophet to model seasonal trends more accurately.

Code:
https://colab.research.google.com/drive/1r1gbfV1AG8y-p3cNX_yYHVoxreiK37cE#scrollTo=2Wz2J0OABDfH

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